

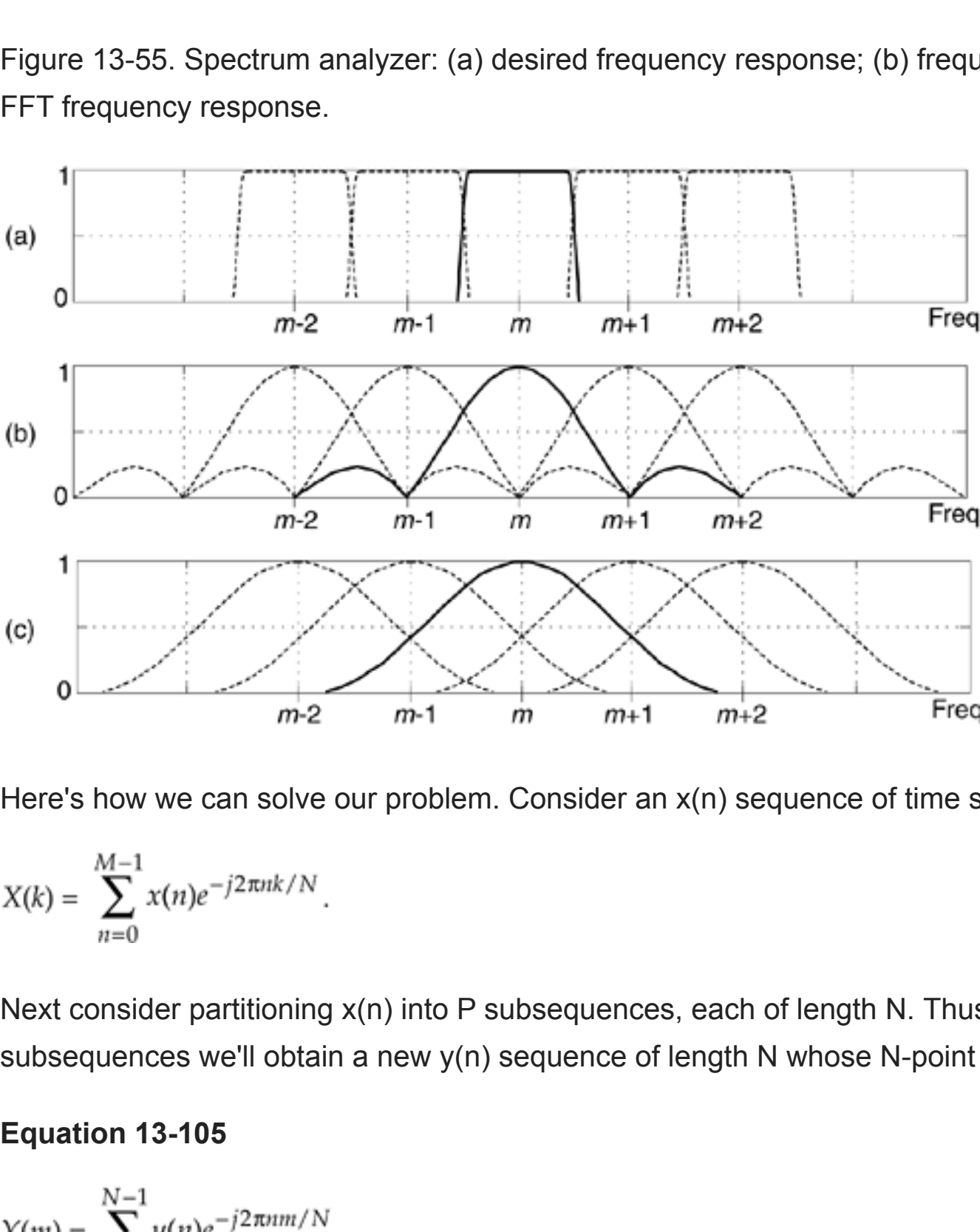
A PRACTICAL SPECTRUM ANALYZER

PREVIOUS PAGE	TABLE OF CONTENT	NEXT PAGE
---------------	------------------	-----------

Here's a clever trick for implementing a practical spectrum analyzer by modifying the time-domain data before applying a radix-2 FFT algorithm.

Let's say we need to build a spectrum analyzer to display, in some manner, the spectral magnitude of a time-domain sequence. We'd like our spectrum analyzer, a bank of bandpass filters, to have a frequency magnitude response something like that shown in Figure 13-55(a). For spectrum analysis, the radix-2 FFT algorithm comes to mind first, as it should. However, the frequency response of individual FFT bins is that shown in Figure 13-55(b), with their non-flat passbands, unpleasantly high sidelobes due to spectral leakage, and overlapped main lobes. We can reduce the leakage sidelobe levels by windowing the time-domain sequence, but that leads to the increased main lobe overlap shown in Figure 13-55(c) and degraded frequency resolution, and we still have considerable droop in the passband response.

Figure 13-55. Spectrum analyzer: (a) desired frequency response; (b) frequency response of standard FFT bins; (c) windowed-data FFT frequency response.



Here's how we can solve our problem. Consider an $x(n)$ sequence of time samples of length M whose M -point DFT is

$$X(k) = \sum_{n=0}^{M-1} x(n)e^{-j2\pi nk/M}$$

Next consider partitioning $x(n)$ into P subsequences, each of length N . Thus $PN = M$. If we add, element for element, the P subsequences we'll obtain a new $y(n)$ sequence of length N whose N -point DFT is

Equation 13-105

$$Y(m) = \sum_{n=0}^{N-1} y(n)e^{-j2\pi mn/N}$$

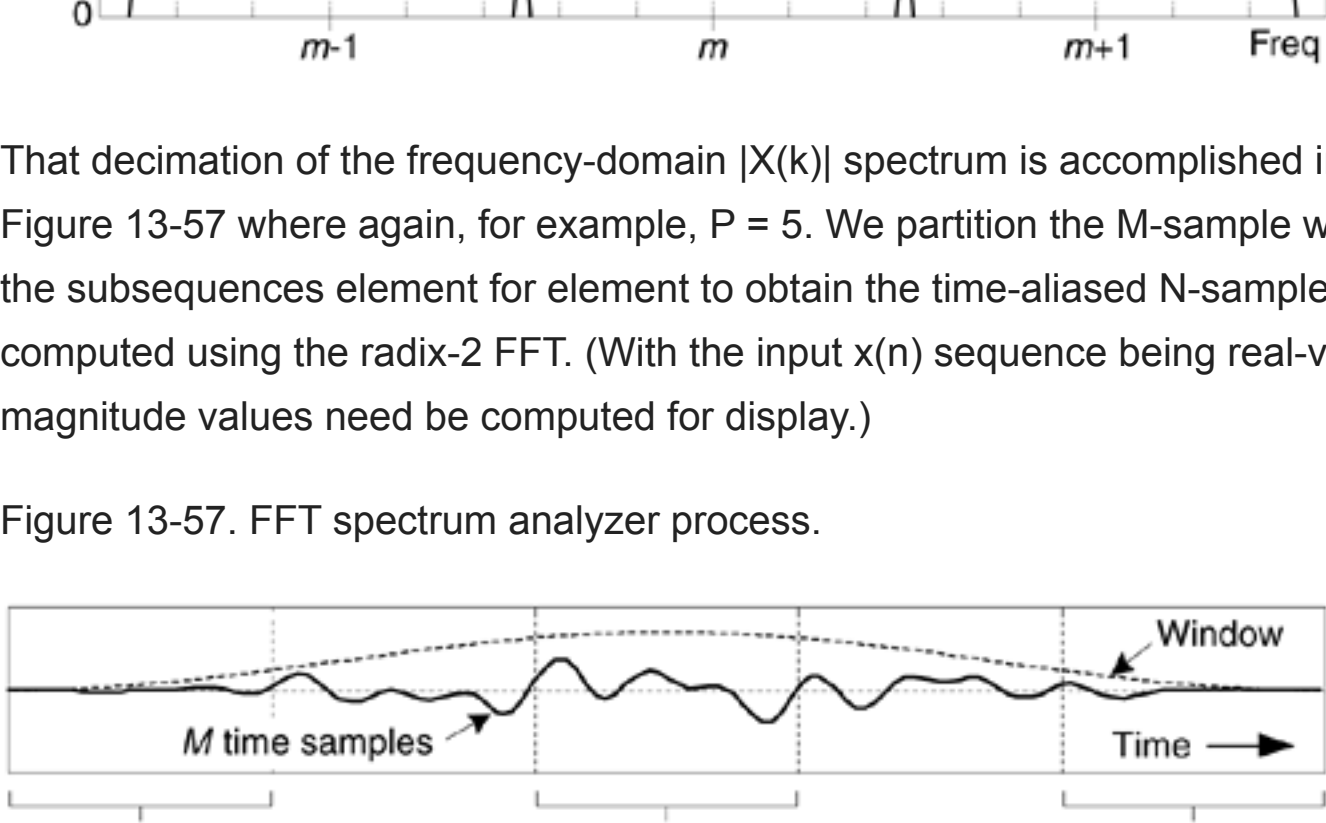
The good news is that

Equation 13-106

$$|Y(m)| = |X(Pm)|$$

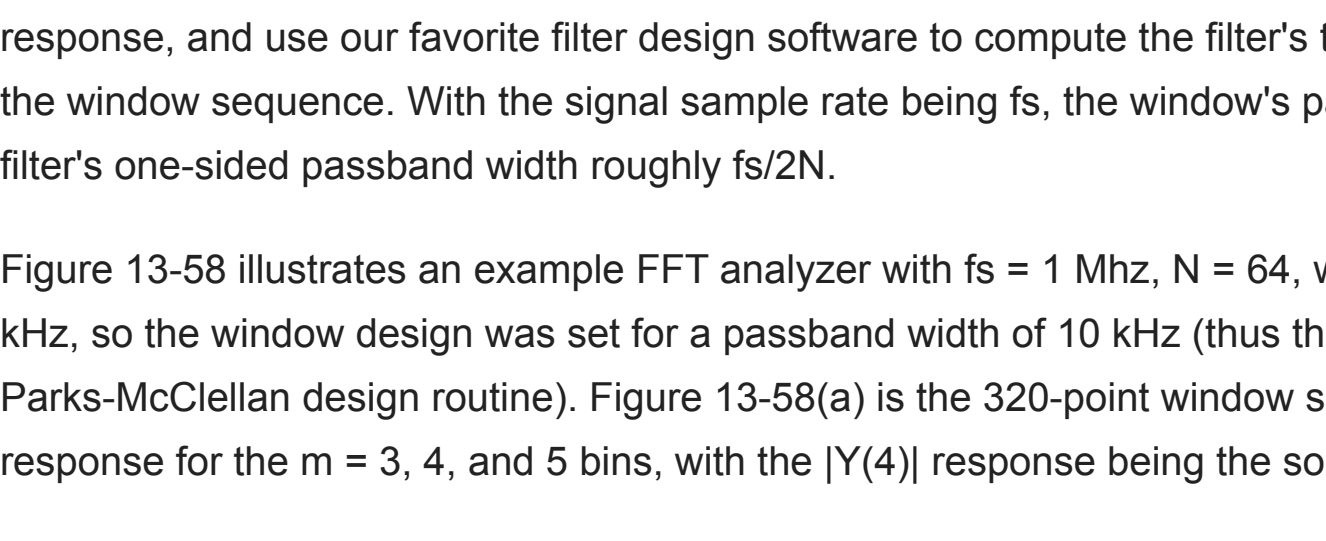
That is, the DFT magnitudes of sequence $y(n)$ are equal to a subset of the longer DFT magnitudes of $x(n)$. $Y(m)$ is equal to a decimated-by- P version of $X(k)$. The relationship between $|Y(m)|$ and $|X(Pm)|$ doesn't seem too important, but here's how we'll take advantage of that equality. We'll create an M -point window sequence whose single-bin frequency response, of an M -point FFT, is the bold curve in Figure 13-56(a). Instead of computing all M FFT outputs, we'll only compute every P th output of the M -point FFT, implementing Eq. (13-105), giving us the decimated FFT bins shown in Figure 13-56(b). In that figure $P = 5$.

Figure 13-56. FFT spectrum analyzer frequency responses.



That decimation of the frequency-domain $|X(k)|$ spectrum is accomplished in the time domain by a time-aliasing operation as shown in Figure 13-57 where again, for example, $P = 5$. We partition the M -sample windowed- $x(n)$ time sequence into $P = 5$ subsequences, sum the subsequences element for element to obtain the time-aliased N -sample $y(n)$ sequence. Next the $|Y(m)|$ spectral magnitudes are computed using the radix-2 FFT. (With the input $x(n)$ sequence being real-valued, with spectral symmetry, only $N/2+1$ the $|Y(m)|$ magnitude values need be computed for display.)

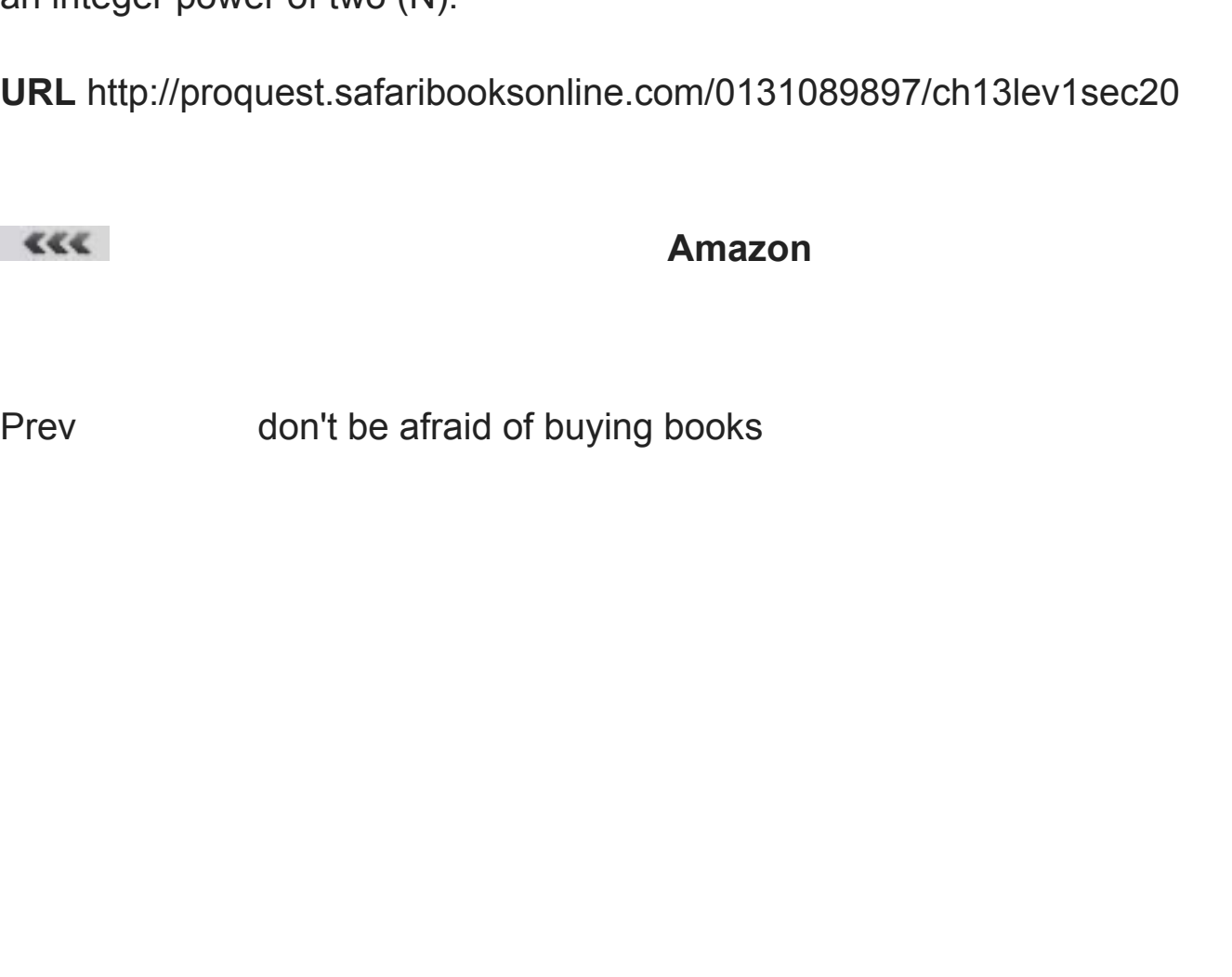
Figure 13-57. FFT spectrum analyzer process.



This process, sweet in its simplicity, is called the weighted overlap-add structure[50,51], and is alternatively referred to as the window-presum FFT[52]. The most difficult part of building this analyzer is designing the M -point window sequence used to window the original $x(n)$ sequence. We do that by specifying the window's frequency domain characteristics just as if it were a digital filter frequency response, and use our favorite filter design software to compute the filter's time-domain impulse response. That impulse response is the window sequence. With the signal sample rate being f_s , the window's passband width will be just less than f_s/N . This makes the filter's one-sided passband width roughly $f_s/2N$.

Figure 13-58 illustrates an example FFT analyzer with $f_s = 1$ Mhz, $N = 64$, with $P = 5$ making $M = 320$. The FFT bin spacing is 15.63 kHz, so the window design was set for a passband width of 10 kHz (thus the filter's one-sided bandwidth was specified as 5 kHz in a Parks-McClellan design routine). Figure 13-58(a) is the 320-point window sequence, while Figure 13-58(b) shows FFT analyzer's response for the $m = 3, 4$, and 5 bins, with the $|Y(4)|$ response being the solid curve.

Figure 13-58. FFT analyzer example: (a) window sequence; (b) analyzer response for 64-point FFT bins $|Y(3)|$, $|Y(4)|$, and $|Y(5)|$.



The width of the spectrum analyzer's passbands is primarily controlled by the window's passband width. The center frequencies of the analyzer's individual passbands is defined by f_s/N . What this means is that the amount of overlap in the analyzer's passbands depends on both the window's passband width, f_s , and N . The dynamic range of the analyzer can be increased by increasing P , which increases M and lengthens the $x(n)$ sequence. As M is increased, the longer window sequence will yield analyzer passbands having a more rectangular shape, lower sidelobes, and reduced passband ripple.

Again, to implement this radix-2 FFT spectrum analyzer, the length of the time-domain sequence (M) must be an integer multiple (P) of an integer power of two (N).

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Prev

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Next

Chapter One. Discrete Sequences and Systems
- Chapter One, Discrete Sequences and Systems - DISCRETE SEQUENCES AND THEIR NOTATION - SIGNAL AMPLITUDE, MAGNITUDE, POWER - SIGNAL PROCESSING OPERATIONAL SYMBOLS - INTRODUCTION TO DISCRETE LINEAR TIME-INVARIANT SYSTEMS - DISCRETE LINEAR SYSTEMS - TIME-INVARIANT SYSTEMS - THE COMMUTATIVE PROPERTY OF LINEAR TIME-INVARIANT SYSTEMS - ANALYZING LINEAR TIME-INVARIANT SYSTEMS - REFERENCES
Chapter Two. Periodic Sampling
- Chapter Two, Periodic Sampling - ALIASING: SIGNAL AMBIGUITY IN THE FREQUENCY DOMAIN - SAMPLING LOW-PASS SIGNALS - SAMPLING BANDPASS SIGNALS - SPECTRAL INVERSION IN BANDPASS SAMPLING - REFERENCES
Chapter Three. The Discrete Fourier Transform
- Chapter Three, The Discrete Fourier Transform - UNDERSTANDING THE DFT EQUATION - DFT SYMMETRY - DFT LINEARITY - DFT MAGNITUDES - DFT FREQUENCY AXIS - DFT SHIFTING THEOREM - INVERSE DFT - DFT LEAKAGE - WINDOWS - DFT SCALLOPING LOSS - DFT RESOLUTION, ZERO PADDING, AND FREQUENCY-DOMAIN SAMPLING - DFT PROCESSING GAIN - THE DFT OF RECTANGULAR FUNCTIONS - THE DFT FREQUENCY RESPONSE TO A COMPLEX INPUT - THE DFT FREQUENCY RESPONSE TO A REAL COSINE INPUT - THE DFT SINGLE-BIN FREQUENCY RESPONSE TO A REAL COSINE INPUT - INTERPRETING THE DFT - REFERENCES
Chapter Four. The Fast Fourier Transform
- Chapter Four, The Fast Fourier Transform - RELATIONSHIP OF THE FFT TO THE DFT - HINTS ON USING FFTS IN PRACTICE - FFT SOFTWARE PROGRAMS - DERIVATION OF THE RADIX-2 FFT ALGORITHM - FFT INPUT/OUTPUT DATA INDEX BIT REVERSAL - RADIX-2 FFT BUTTERFLY STRUCTURES - REFERENCES
Chapter Five. Finite Impulse Response Filters
- Chapter Five, Finite Impulse Response Filters - AN INTRODUCTION TO FINITE IMPULSE RESPONSE (FIR) FILTERS - CONVOLUTION IN FIR FILTERS - LOW-PASS FIR FILTER DESIGN - HIGH-PASS FIR FILTER DESIGN - REMOVAL EXCHANGE FIR FILTER DESIGN METHOD - HALF-BAND FIR FILTERS - PHASE RESPONSE OF FIR FILTERS - A GENERIC DESCRIPTION OF DISCRETE CONVOLUTION - REFERENCES
Chapter Six. Infinite Impulse Response Filters
- Chapter Six, Infinite Impulse Response Filters - AN INTRODUCTION TO INFINITE IMPULSE RESPONSE FILTERS - THE LAPLACE TRANSFORM - THE Z TRANSFORM - IMPULSE INVARIANCE IIR FILTER DESIGN METHOD - BILINEAR TRANSFORM IIR FILTER DESIGN METHOD - OPTIMIZED IIR FILTER DESIGN METHOD - PITFALLS IN BUILDING IIR FILTERS - IMPROVING IIR FILTERS WITH CASCADED STRUCTURES - A BRIEF COMPARISON OF IIR AND FIR FILTERS - REFERENCES
Chapter Seven. Specialized Lowpass FIR Filters
- Chapter Seven, Specialized Lowpass FIR Filters - FREQUENCY SAMPLING FILTERS: THE LOST ART - INTERPOLATED LOWPASS FIR FILTERS - REFERENCES
Chapter Eight. Quadrature Signals
- Chapter Eight, Quadrature Signals - WHY CARE ABOUT QUADRATURE SIGNALS? - THE NOTATION OF COMPLEX NUMBERS - REPRESENTING REAL SIGNALS USING COMPLEX PHASORS - A FEW THOUGHTS ON NEGATIVE FREQUENCY - QUADRATURE SIGNALS IN THE FREQUENCY DOMAIN - BANDPASS QUADRATURE SIGNALS IN THE FREQUENCY DOMAIN - COMPLEX DOWN-CONVERSION - A COMPLEX DOWN-CONVERSION EXAMPLE - AN ALTERNATE DOWN-CONVERSION METHOD - REFERENCES
Chapter Nine. The Discrete Hilbert Transform
- Chapter Nine, The Discrete Hilbert Transform - HILBERT TRANSFORM DEFINITION - WHY CARE ABOUT THE HILBERT TRANSFORM? - IMPULSE RESPONSE OF A HILBERT TRANSFORMER - DESIGNING A DISCRETE HILBERT TRANSFORMER - TIME-DOMAIN ANALYTIC SIGNAL GENERATION - COMPARING ANALYTIC SIGNAL GENERATION METHODS - REFERENCES
Chapter Ten. Sample Rate Conversion
- Chapter Ten, Sample Rate Conversion - DECIMATION - INTERPOLATION - COMBINING DECIMATION AND INTERPOLATION - POLYPHASE FILTERS - CASCADED INTEGRATOR-COMB FILTERS - REFERENCES
Chapter Eleven. Signal Averaging
- Chapter Eleven, Signal Averaging - COHERENT AVERAGING - INCOHERENT AVERAGING - AVERAGING MULTIPLE FAST FOURIER TRANSFORMS - FILTERING ASPECTS OF TIME-DOMAIN AVERAGING - EXPONENTIAL AVERAGING - REFERENCES
Chapter Twelve. Digital Data Formats and Their Effects
- Chapter Twelve, Digital Data Formats and Their Effects - FIXED-POINT BINARY FORMATS - BINARY NUMBER PRECISION AND DYNAMIC RANGE - EFFECTS OF FINITE FIXED-POINT BINARY WORD LENGTH - FLOATING-POINT BINARY FORMATS - BLOCK FLOATING-POINT BINARY FORMAT - REFERENCES
Chapter Thirteen. Digital Signal Processing Tricks
- Chapter Thirteen, Digital Signal Processing Tricks - FREQUENCY TRANSLATION WITHOUT MULTIPLICATION - HIGH-SPEED VECTOR MAGNITUDE APPROXIMATION - FREQUENCY-DOMAIN WINDOWING - FAST MULTIPLICATION OF COMPLEX NUMBERS - EFFICIENTLY PERFORMING THE FFT OF REAL SEQUENCES - COMPUTING THE INVERSE FFT USING THE FORWARD FFT - SIMPLIFIED FIR FILTER STRUCTURE - REDUCING A/D CONVERTER QUANTIZATION NOISE - A/D CONVERTER TESTING TECHNIQUES - FAST FIR FILTERING USING THE FFT - GENERATING NORMALLY DISTRIBUTED RANDOM DATA - ZERO-PHASE FILTERING - SHARPENED FIR FILTERS - INTERPOLATING A BANDPASS SIGNAL - SPECTRAL PEAK LOCATION ALGORITHM - COMPUTING FFT TWIDDLE FACTORS - SINGLE-TONE DETECTION - THE SLIDING DFT - THE ZOOM FFT - A PRACTICAL SPECTRUM ANALYZER - AN EFFICIENT ARC TANGENT APPROXIMATION - FREQUENCY DEMODULATION ALGORITHMS - DC REMOVAL - IMPROVING TRADITIONAL CIC FILTERS - SMOOTHING IMPULSIVE NOISE - EFFICIENT POLYNOMIAL EVALUATION - DESIGNING VERY-HIGH-ORDER FIR FILTERS - TIME-DOMAIN INTERPOLATION USING THE FFT - FREQUENCY INTERPOLATION USING DECIMATION - AUTOMATIC GAIN CONTROL (AGC) - APPROXIMATE ENVELOPE DETECTION - A QUADRATURE OSCILLATOR - DUAL-MODE AVERAGING - REFERENCES
Appendix A. The Arithmetic of Complex Numbers
- Appendix A, The Arithmetic of Complex Numbers - Section A.1. GRAPHICAL REPRESENTATION OF REAL AND COMPLEX NUMBERS - Section A.2. ARITHMETIC REPRESENTATION OF COMPLEX NUMBERS - Section A.3. ARITHMETIC OPERATIONS OF COMPLEX NUMBERS - Section A.4. SOME PRACTICAL IMPLICATIONS OF USING COMPLEX NUMBERS - REFERENCES
Appendix B. Closed Form of a Geometric Series
Appendix B, Closed Form of a Geometric Series
Appendix C. Time Reversal and the DFT
Appendix C, Time Reversal and the DFT
Appendix D. Mean, Variance, and Standard Deviation
- Appendix D, Mean, Variance, and Standard Deviation - Section D.1. STATISTICAL MEASURES - Section D.2. STANDARD DEVIATION, OR RMS, OF A CONTINUOUS SINEWAVE - Section D.3. THE MEAN AND VARIANCE OF RANDOM FUNCTIONS - Section D.4. THE NORMAL PROBABILITY DENSITY FUNCTION - REFERENCES
Appendix E. Decibels (dB and dBm)
- Appendix E, Decibels (dB and dBm) - Section E.1. USING LOGARITHMS TO DETERMINE RELATIVE SIGNAL POWER - Section E.2. SOME USEFUL DECIBEL NUMBERS - Section E.3. ABSOLUTE POWER USING DECIBELS
Appendix F. Digital Filter Terminology
- Appendix F, Digital Filter Terminology - REFERENCES
Appendix G. Frequency Sampling Filter Derivations
- Appendix G, Frequency Sampling Filter Derivations - Section G.1. FREQUENCY RESPONSE OF A COMB FILTER - Section G.2. SINGLE COMPLEX FSF FREQUENCY RESPONSE - Section G.3. MULTISECTION COMPLEX FSF PHASE FREQUENCY RESPONSE - Section G.4. MULTISECTION COMPLEX FSF FREQUENCY RESPONSE - Section G.5. REAL FSF TRANSFER FUNCTION - Section G.6. TYPE-IV FSF FREQUENCY RESPONSE
Appendix H. Frequency Sampling Filter Design Tables
Appendix H, Frequency Sampling Filter Design Tables

PREVIOUS PAGE	TABLE OF CONTENT	NEXT PAGE
---------------	------------------	-----------



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